Shakou Kakga

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Introduction

This illustration concerns a variety of Hakka (iso 639:hak) spoken in Shakou Township, Yingde County-level City, in Guangdong's Qingyuan Prefecture. Hakka (客家, Standard Mandarin *Kèjiā*) is a group of Sinitic varieties with substantial internal diversity. This illustration aims to provide a picture of this internal diversity, representing the Hakka varieties of Northern Guangdong, which are distinct in a number of important respects from Meixian Hakka, previously illustrated by Lee and Zee (2009). In fact, the variety under discussion here fails to exhibit many of the features singled out by Norman (1989) as diagnostic of a Hakka dialect, though its speakers clearly identify it as such. To avoid confusion with other Hakka varieties, we have used the local endonym *Kakga* /kʰak˦ .ka˦ / for the name of the language. Our consultant reports a high level of mutual intelligibility between Shakou Kakga and other Hakka varieties of Northern Guangdong, but more limited mutual intelligibility with Meixian Hakka and in particular Taiwanese Hakka varieties. Shakou Kakga is the language of daily communication in Shakou township, existing in a triglossic relationship with Cantonese (used as a *lingua franca* in Northern Guangdong) and Standard Mandarin (as the official language of the PRC).

Previous work on Hakka varieties spoken in Yingde County-level City include two short phonological sketches and wordlists in Chinese (Hu 2003, Qiu and Li 2007), with minimal discussion of the phonetic realisation of the identified categories. To our knowledge, this is the first work on the Shakou subvariety.

Most of the examples presented here are isolated roots, represented in the Chinese writing system as a single character. These are convenient to elicit and provide a good range of minimal contrasts. However, they may not always be able to stand alone in ordinary speech as fully-fledged words, and the meanings of the roots may depend greatly on the context – the glosses here are only approximate, giving one of a number of possible interpretations. Words in Sinitic languages such as Shakou Kagka are typically composed of a combination of these roots: where pronunciation in isolation differs from pronunciation in combination, this has been noted.

In addition to IPA transcription we give Chinese characters for Hakka forms where possible. The forms given are generally those used for cognate items in Mandarin or (if an item is not present in Mandarin) in Cantonese, which, as mentioned, is used as a local lingua franca in the area where Shakou Kagka is spoken. Where an item has no written form we indicate this with □. We have used the traditional, rather than simplified, forms of Chinese characters, partly as a matter of familiarity to the authors, though it should be noted that these forms will be used only by the very oldest speakers of the language.

1 Consonants

All Shakou Kakga syllables are of the form (C)VC or (C)VV. Words are typically composed of one or more monosyllabic roots of this type. The permissible consonants in syllable-initial and syllable-final position are quite distinct, with 21 contrastive consonants in syllable initial position as compared to four in syllable-final position.

1.1 Syllable-initial Consonants

	Labial	**Alveolar**	**Postalveolar**	**Velar**	**Labialized Velar**	**Glottal**
Plosive	p^h p		t^h t	k^h k	k^{hw} k^w	
Affricate		ts^h ts	$tʃ^h$ tʃ			
Nasal	m			ŋ		
Fricative	f	s	ʃ			h
Approximant			j		w	
Lateral approximant		l				

伯 pak˧˩ ‘uncle’	打 ta˧˨ ‘hit’	交 kau˥ ‘exchange’
白 pʰak˥ ‘white’	鄧 tʰən˧˨ (surname)	家 kʰa˥ ‘family’
馬 ma˥ ‘horse’	嗱 la˧˨ ‘there!’	鴉 ŋa˥ ‘crow’
花 fa˥ ‘flower’	沙 sa˥ ‘sand’	下 ha˥ ‘down’
横 waŋ˩˧ ‘horizontal’	蛇 ʃa˩˧ ‘snake	瓜 kʷa˥ ‘melon’
	責 tsak˧˩ ‘debt’	誇 kʰʷa˥ ‘exaggerate’
	茶 tsʰa˩˧ ‘tea’[1]	野 ja˥ ‘wild’
	展 tʃən˧˨ ‘display’	
	車 tʃʰa˥ ‘car’	

Syllable-initial plosives show a two-way contrast for aspiration. There is no contrast for voicing, with three caveats. Firstly, unaspirated stops are occasionally realised with some degree of voicing in isolation and in connected speech. Secondly, in connected speech, unvoiced segments may undergo voicing in voiced environments. Thirdly, initial nasals undergo partial denasalisation, resulting in the segments being realised as prenasalised voiced stops [ᵑɡ, ᵐb]. The degree to which this process applies depends to some extent on speech rate, and in slow speech the nasalisation may be lost entirely, resulting in a secondary voicing contrast. This would distinguish the second person pronoun 你 /ŋi˧˨ / [(ᵑ)ɡi˧˨] from the third person pronoun 其 /ki˧˨ / [ki˧˨].

This denasalisation process forms part of a wider set of ‘obstruentisation’ processes which apply to all initial sonorants in Shakou Kakga, including a centralisation process of initial /l/ before high vowels (so that 脷 /li˧˨ / ‘tongue’ is realised as [ˡdi˧˨]). The glides /j/ and /w/ undergo frication before high vowels, so that the root 醫 /ju˥ / ‘medical’ is realised [ʝɨ˥],[2] while 烏 /wu˥ / ‘black’ is realised with a bilabial fricative as [βɵ˥].

In contrast to Meixian Hakka, Shakou Kakga preserves a distinction between alveolar and postalveolar fricatives and affricates. Postalveolars have a palato-alveolar realisation, readily distinguishable from e.g. Mandarin Chinese alveolo-palatals and retroflexes.

1.2 Syllable-final consonants

	Non-Velar	Velar
Plosive	t	k
Nasal	n	ŋ

[1]This item is exemplified in the compound 茶水 /tsʰa˩˧ ʃui˧˨ / ‘tea-water’.
[2]This root is also realised as /ji˥ / [ji˥] in some contexts.

甩 lot˦ 'flee'	落 lok˦ 'descend'
亂 lon˧ 'disordered'	狼 loŋ˧˩ 'wolf'

Syllable-final plosives are unreleased and realised with some glottal constriction. The 'non-velar' stops here surface with a wide range of place realisations. Final /t/ and /n/ assimilate to following labial place, so that 發明 /fat.min/ 'invent' is realised as [fɑp̚.ᵐbin], and 錢包 /tsʰyn.bau/ 'wallet' as [tsʰiɛm.baʊ]. Before syllables which do not begin with non-continuant consonants, /t/ is typically realised as a glottal stop, as in 北歐 /bət.əu/ [bəʔ.ɛʉ] 'Northern Europe'. In isolation our primary consultant generally articulates these stops as doubly articulated alveolar-labials [t͡p, n͡m], but some other speakers produce simple alveolars [t, n]. For simplicity of representation we consistently represent these stops as /t, n/.

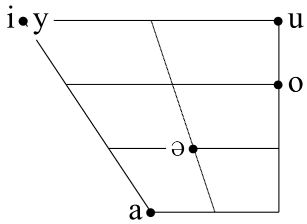
The oral constriction of both non-velar and velar stops is often lost, in which case they are distinguished by vowel quality – as detailed below, all vowels show distinct allophonic alternants before underlying velar and non-velar stops.

2 Vowels

Providing a phonemic representation of vowels in Sinitic languages presents a problem – the inventory of vowels phones may be quite different in different consonantal environments. We also typically lack alternations between complementary vowel phones, as a consequence of the relatively simple morphology in these languages. What this means is that it is difficult to be certain whether to analyse two complementary vowel phones as realisations of the same phoneme conditioned by different environments, or as different phonemes phonotactically constrained to appear in particular environments. This is compounded by the fact that any given vowel phone will be complementary not only with one other phone but many, so that complementarity alone cannot be used as a criterion to identify alternants of a single phoneme.

We have chosen to minimise the size of the phonemic inventory, attempting to analyse vowel phones which appear in complementary environments as allophones where possible. In general, we have used phonetic similarity as our main heuristic when deciding whether two complementary phones are allophones of a single phoneme. This means, however, that the actual phonetic realisation of a phone may be quite different to its phonemic representation – for example, /y/ preceding a non-velar final is realised as [iɛ].

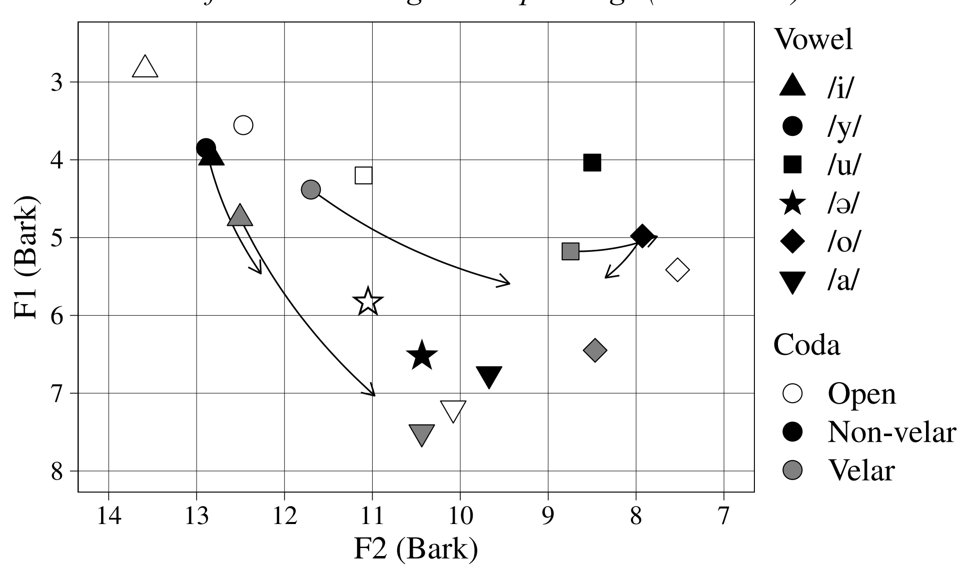
2.1 Monophthongs



嗱 la˦˨ ‘there!; take it!’	辣 lat˦ ‘spicy’	(客 kʰak˨˩ ‘guest’)
脷 li˦˨ ‘tongue’	力 lit˦ ‘power’	叻 lik˦ ‘clever’
路 lu˦˨ ‘road’	律 lut˦ ‘law’	六 luk˨˩ ‘six’
老 lo˦˨ ‘old’	甩 lot˦ ‘flee’	落 lok˦ ‘descend’
旅 ly˦ ‘travel’	列 lyt˦ ‘row’	略 lyk˦ ‘omit’
呢 lə˦˨ (final particle)[3]	竻 lət˨˩ ‘thorn’	

Shakou Kakga shows at most 6 vowel quality contrasts in any environment. The realisation of the contrasts varies substantially between open and closed syllables on the one hand, and syllables closed by velar and non-velar segments on the other. One of these contrasts (between /a/ and /ə/) is largely absent in open syllables, as well as syllables with velar codas, such that only /a/ is usually found, with /ə/ being limited to a small set of functional items. We will deal with each of these environments in turn.

(1) *Formant Values for Shakou Kakga Monophthongs (Bark scale)*



[3]Exemplified in the phrase 接下來呢 /tsyt˦ ha˦ loi˨˦ lə˦˨ / ‘and then...’

Open Syllables In open syllables, /u/ is generally centralised, being realised in most environments as [ɵ]. After alveolar fricatives and affricates, this is fronted and unrounded to [ɨ], sometimes being realised as a syllabic fricative [z̩].[4] This realisation is also found following a palatal glide, where all instances of /u/ may also be pronounced with /i/ (though not vice-versa).[5] /o/ has a somewhat lowered realisation [o̞]. /ə/, which has a somewhat fronted realisation [ə̟], occurs in open syllables only in a limited class of functional items. Our primary consultant consistently keeps /y/ and /u/ distinct, but they appear to be merged (as [y]) by a number of other speakers in open syllables.

(2) *Vowels in open syllables*

Character	**Phonemic**	**Broad Phonetic**	**Gloss**
喥	/la˧˨ /	[la˧˨]	‘there!; take it!’
脷	/li˧˨ /	[li˧˨]	‘tongue’
路	/lu˧˨ /	[lɵ˧˨]	‘road’
老	/lo˧˨ /	[lo̞˧˨]	‘old’
旅	/ly˧ /	[ly˧]	‘travel’
了	/lə˧˨ /	[lə˧˨]	(aspect marker)
數	/su˥˧ /	[sɨ˥˧]∼[sz̩˥˧]	‘number’

Closed Syllables (Velar) High front vowels are lowered and diphthongised before velar codas, so that /i/ is realised as [ɪa] and /y/ as [ʏɔ]. The high back vowel /u/ is lowered to [o]. /o/ is lowered and somewhat unrounded to to [ɔ̜]. The phoneme /ə/ does not occur before velar codas.

(3) *Vowels before velar codas*

Character	**Phonemic**	**Broad Phonetic**	**Gloss**
客	/kʰak˨˩ /	[kʰak˨˩]	‘guest’
叻	/lik˧ /	[lɪak˧]	‘clever’
六	/luk˨˩ /	[lok˨˩]	‘six’
落	/lok˧ /	[lɔk˧]	‘descend’
略	/lyk˧ /	[lʏɔk˧]	‘omit’

[4] We analyse this as an instance of /u/ rather than /i/ because we see a constrast with items like 西 /si˧ / ‘west’, but we see no corresponding contrast with a nonlow rounded vowel.

[5] Examples of this include 醫 /ju˧ /∼/ji˧ / ‘medicine’ and the homophonous 衣 /ju˧ /∼/ji˧ / ‘clothing’.

Closed Syllables (Non-velar) Before non-velar codas, /i/ is somewhat centralised and variably diphthongised [ɪ(i)]. The front rounded vowel /y/ is unrounded and diphthongised [iɛ]. The back rounded vowels are centralised and (variably in the case of /u/) diphthongised, so that /o/ surfaces as [ɔ̈ə] and /u/ as [ʊ(ə)]. The low vowel /a/ has a back realisation [ɑ], while /ə/ is somewhat lowered [ə̞].

(4) *Vowels before non-velar codas*

Character	**Phonemic**	**Broad Phonetic**	**Gloss**
辣	/lat˧ /	[lɑt˧]	‘spicy’
力	/lit˧ /	[lɪt˧]	‘power’
律	/lut˧ /	[lʊt˧]	‘law’
甩	/lot˧ /	[lɔət˧]	‘flee’
列	/lyt˧ /	[liɛt˧]	‘row’
竻	/lət˧˩ /	[lət˧˩]	‘thorn’

2.2 Diphthongs

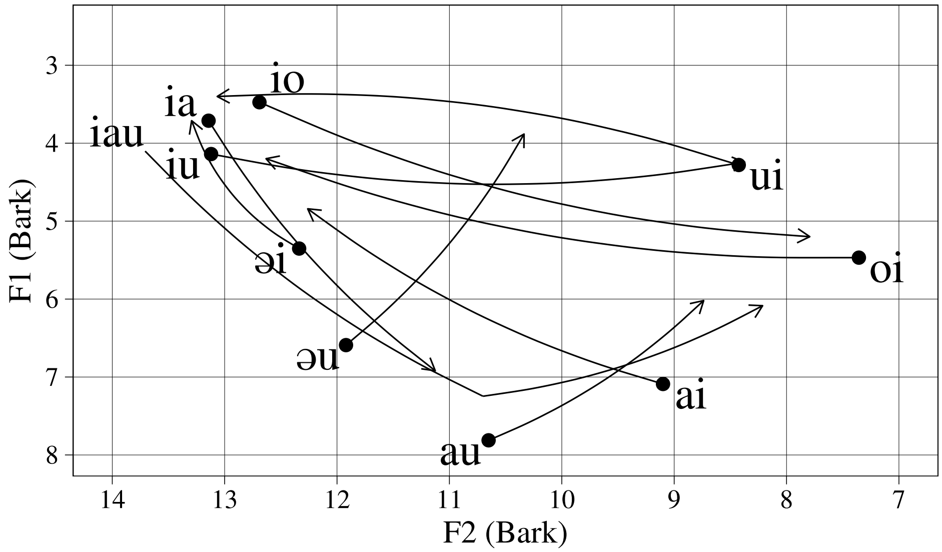
Primary diphthongs are found only in open syllables, although (as discussed above), diphthongisation processes in closed syllables may mean that additional diphthongs occur as allophones of monophthongal phonemes.

We analyse all primary diphthongs in Shakou Kakga as being composed of a monophthongal vowel phoneme (except y) and one of the phonemes /i, u/. We represent the occurring diphthongs as /iu, ia, io, ui, əi, əu, oi, ai, au/ (/ou/ does not occur). The respective phonetic realisations of these diphthongs are [ɪʊ, ia, io, ʊi, ɛʉ, ei, oi, ɑi, ao]. We also observe a triphthong /iau/ [iao].

(5) *Shakou Kagka dipthongs*

拉 lai˧ ‘pull’	鬧 lau˥˧ ‘noisy’
雷 lui˩˧ ‘thunder’	留 liu˩˧ ‘remain’
來 loi˩˧ ‘come’	料 liau˥˧ ‘material’
泥 ləi˩˧ ‘mud’	樓 ləu˩˧ ‘building’

(6) *Formant Values for Shakou Kakga Diphthongs (Bark scale)*



The diphthongs /ia, io/ appear only after alveolar affricates and velars.

嘢 ŋia˧ ‘thing’	□ gio˧ ‘run around’

These interactions with onsets, together with the fact that the initial element in these diphthongs is very short, might lead us to conclude that these forms should be analysed as containing a glide-vowel pair. However, this would suggest that these sequences should be permitted in closed syllables, which we do not observe.[6]

3 Tones

There are four contrastive tones in Shakou Kakga. In obstruent-closed syllables we see only a two-way contrast. A minimal quadruplet can be seen with the segmental form /mun/.

[6]We DO see similar (though not identical) phonetic sequences appearing before velar codas, but these are analysed as allophones of monophthongs – these sequences never appear before non-velar codas.

(7) *Shakou Kagka tone contrasts [non-checked syllables]*

蚊 mun˧ ‘mosquito’
門 mun˩˧ ‘door’
悶 mun˨ ‘stuffy’
問 mun˥˧ ‘ask’

These tones may appear in all syllable types without codas or with nasal codas, with the exception that we do not observe the low fall-rise tone following unaspirated stops.[7]

Syllables closed with plosive codas (‘checked’ syllables) only show a simple high-low distinction. A minimal pair can be seen with the segmental form /ŋit/. The tones are essentially identical to the high and low falling tones in unchecked syllables, except that the low fall in checked syllables appears to be somewhat steeper.

(8) *Shakou Kagka tone contrasts [checked syllables]*

日 ŋit˨˩ ‘day’
入 ŋit˧ ‘enter’

We have been unable to detect any phonological tone sandhi phenomena in Shakou Kagka. Various instances of tonal reduction and tonal co-articulation effects in fast speech can be observed in the example text, but we are unable to provide a full account of these here.

4 Ideophones

In common with other Sinitic languages (and languages more generally), Shakou Kakga has a well-defined class of IDEOPHONES. Ideophones are marked words that depict sensory imagery (Dingemanse 2012, Akita & Dingemanse 2019), and are known to exhibit sound patterns not found in prosaic words (Diffloth 1979; Childs 1988; Akita et al. 2013; Thompson & Do 2019; cf. Thompson 2018 for Chinese languages). We do indeed find that Shakou Kakga ideophones display segmental properties distinct from non-ideophonic words.

For example, we see a tap [ɾ] in the ideophone [kuɾukuɾu] ‘(sound of) guzzling’. We also see a number of prominent fricative vowels in ideophones, as for example in [vv̩vv̩]

[7] This is for historical reasons – this tonal category was historically restricted to occuring with VOICED stops, which subsequently became aspirated stops.

‘(sound of) infant crying’ and [dzz̩z̩] ‘(sound of) electricity crackling’. This last example also displays a voiced fricative onset which is not found in non-ideophonic words.

As Thompson (2018) observes in multiple Sinitic varieties, ideophonic words show a distinct distribution of tones, and this also holds for Shakou Kakga. The constituent syllables of ideophones in Shakou Khakga show an apparent preference for a high level tone /˥/ specification, though there are various exceptions, e.g. /put˧˩/ ‘(sound of) farting’, /k^{h}ok˧˩ k^{h}ok˧˩ k^{h}o˧˨ ka˨˩˨/ ‘(sound of) cock crowing’.

Note that the Shakou Kakga ideophones reported here phonologically differ from their synonymous counterparts reported by Mok (2001) in her detailed study on the phonological structure of ideophones from Sixian Hakka spoken in Taiwan.

5 Transcription of recorded passage

有一次北風同太陽正在爭蠻人比較有本事。
jiu˥ it˥ ts^{h}u˥˧ pət˥ fuŋ˥ t^{h}uŋ˧˨ t^{h}ai˥˧ .joŋ˨˩˨ ʧin˥ .ts^{h}oi˧˨ tsaŋ˧˨ man˧˨ .ŋin˨˩˨ pi˧˨ .kau˥˧ jiu˥ pun˧˨ .su˥˧
“Once, the north wind and the sun were fighting over who was more capable.”

其哩啱好看到有一隻人行過。
ki˧˨ .li˥˧ ŋan˥ .ho˧˨ k^{h}on˥˧ .to˥˧ jiu˥ it˥ tsət˧˩ ŋin˨˩˨ haŋ˨˩˨ .ko˥˧
“Just then, they saw a person walking by.”

個隻人著穩一件斗篷。
kai˥˧ tsət˧˩ ŋin˨˩˨ ʧuk˧˩ wun˨˩˨ it˥ k^{h}yn˧˨ təu˨˩˨ .p^{h}aŋ˧˨
“That person was wearing a cloak’.’

其哩就話「蠻人可以令個隻人脫得個件斗篷就算蠻人比較有本事。」
ki˧˨ .li˥˧ ʧ^{h}iu˧˨ wa˧˨ man˧˨ .ŋin˨˩˨ k^{h}o˧˨ .ji˥ liŋ˥˧ tsət˧˩ ŋin˨˩˨ t^{h}ot˥ tət˧˩ kai˥˧ k^{h}yn˧˨ təu˨˩˨ .p^{h}aŋ˧˨ ʧ^{h}iu˧˨ son˥˧ man˧˨ .ŋin˨˩˨ pi˧˨ .kau˥˧ jiu˥ pun˧˨ .su˧˨ ’
“Then they said ‘Whoever can make the person take off that cloak will be considered the more capable.’ ”

然後呢北風就搏命吹量知得。
jən˨˩˨ .həu˥˧ lə˧˨ pət˥ fuŋ˥ ʧ^{h}iu˧˨ pok˧˩ .miŋ˥˧ ʧ^{h}ui˥ lyŋ˥˧ ti˥ .tət˧˩

“Then the north wind blew as hard as he could.”

其吹得越犀利, 個隻人就將其嘅斗篷抱得越緊。
ki˧ ʧʰui˥ tət˧˩ jət˧˩ səi˩˧ .li˥˧ kai˥˧ tsət˧˩ ŋin˩˧ ʧʰiu˧ tsyŋ˥ ki˧ .kə˥ təu˩˧ .pʰaŋ˧ pau˥ tət˧˩ jət˧˩ kin˧
“The harder he blew, the tighter the person held their cloak.”

最後奮了北風就無辦法了。
tsui˥˧ .həu˥˧ mui˧ lə˧ pət˥ fuŋ˥˧ ʧʰiu˧ mou˧ pʰaŋ˧ .fat˧˩ lə˧
“He could not [succeed] in the end, the north wind could do nothing [else].”

只好放棄。
tsu˧ .ho˧ foŋ˥˧ .kʰi˥˧
“He could only give up.”

接下來呢太陽就出來曬了一下子。
tsyt˥ .ha˥ .loi˩˧ lə˧ tʰai˥ .joŋ˩˧ ʧʰiu˧ ʧʰut˥ .loi˩˧ sai˥ lə˧ it˥ .ha˥ .tsu˧
“Then, the sun came straight out.”

個隻人就馬上脫得其件斗篷。
kai˥˧ tsət˧˩ ŋin˩˧ ʧʰiu˧ ma˥ .ʃoŋ˥˧ tʰot tət˥ ki˧ kʰyn˧ təu˩˧ .pʰaŋ˧
“The person immediately took off their cloak.”

然後北風就只能夠認輸了。
jən˩˧ .həu˥˧ pət˥ fuŋ˥ ʧʰiu˧ tsu˧ lən˩˧ .kəu˥˧ ŋin˧ ʃy˧ lə˧
“Then the north wind could only concede defeat.”

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